数理方程部份.

1. 数学物理方程.

波动方程.

$$\frac{\partial^2 u}{\partial t^2} - a^2 \nabla^2 u = f.$$

直角坐标系中,

$$\frac{\partial^2 u}{\partial t^2} - a^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = f.$$

输运方程.

$$\frac{\partial u}{\partial t} - a^2 \nabla^2 u = f$$

稳定场方程.(Poisson 方程)

$$\nabla^2 u = f.$$

当 $f=0$. 有 Laplace 方程.

$$\nabla^2 u = 0.$$

2. 定解问题.

- 微分方程
- 定解条件
  - 初始条件 ← $t=0$
  - 边界条件. ← $x=0$, $x=l$.

	初始	边界		
波动	$u\|_{t=0}=\varphi(x)$ $\frac{\partial u}{\partial t}\|_{t=0}=\psi(x)$	①	$u\|_{x=0}=A \quad u\|_{x=l}=B$	
		②	$\frac{\partial u}{\partial x}\|_{x=l}=\frac{1}{ES}F(t)$	受力.
输运	$u\|_{t=0}=\varphi(r)$ $r\in\partial\Omega$	①	$u\|_{\partial\Omega}=u_0(r,t),\ r\in\partial\Omega$	环境温
		②	$\frac{\partial u}{\partial n}\|_{\partial\Omega}=\frac{1}{k}q(r,t),\ r\in\partial\Omega$	热流.
		③	$(\frac{\partial u}{\partial n}+hu)\|_{\partial\Omega}=hu_0(r,t) \quad r\in\partial\Omega$	边界热交换
稳定场	与前类似.			

一维定解问题求解.

① 波动方程 第一类边界条件.

例: $\frac{\partial^2 u}{\partial t^2} - 9\frac{\partial^2 u}{\partial x^2} = 0$

$u(0,t) = 0, \quad u(5,t) = 0 \quad t \geq 0.$

$u|_{t=0} = 4\sin(\pi x) - \sin(2\pi x) - 3\sin(5\pi x), \quad \frac{\partial u}{\partial t}|_{t=0} = 0., \quad 0 \leq x \leq 5.$

解: ① 考虑分离变量特殊解 $u(x,t) = X(x)T(t)$.

代入, 有

$$X(x)T''(t) - 9X''(x)T(t) = 0$$

$$\Leftrightarrow \quad \frac{9X''(x)}{X(x)} = \frac{T''(t)}{T(t)}$$ 形式上, 左只与x有关, 右只与t相关.

$$\therefore \quad \frac{X''}{X} = \frac{T''}{9T} = -\lambda$$

$$\begin{cases} \underbrace{X'' + \lambda X = 0} \\ T'' + 9\lambda T = 0. \end{cases}$$

② 约化边界条件:

$$X(0) = X(5) = 0.$$

对 $9X'' + \lambda X = 0$, 考虑解 $A\sin\mu x + B\cos\mu x$

$-\omega^2 + \lambda = 0$, $\omega = \lambda^{\frac{1}{2}}$.

$X(0) = B = 0$, $X(5) = \sin(5\lambda^{\frac{1}{2}}) = 0.$ $5\lambda^{\frac{1}{2}} = n\pi$, $n \in N.$

$\lambda_n = (\frac{1}{5}n\pi)^2.$ $\omega_n = \sqrt{\lambda_n} = \frac{1}{5}n\pi$

$X_n = \sin\frac{1}{5}n\pi x$, $n \in N.$

③ 将 $\lambda_n = (\frac{1}{5}n\pi)^2$ 代入. $T'' + 9\lambda T = 0$, 有 $T'' + (\frac{3}{5}n\pi)^2 T = 0$.

同样考虑, 形如 $(C\sin\mu t + D\cos\mu t)$ 的解.

$$[-v^2 + (\frac{3}{5}n\pi)^2] = 0 \quad \Rightarrow \quad v = \frac{3}{5}n\pi.$$

$$T = C_n \sin\frac{3}{5}n\pi t + D_n \cos\frac{3}{5}n\pi t.$$

$$u = TX = \sum_{n=1}^{\infty} \sin\frac{n\pi x}{5} \left(C_n \sin\frac{3}{5}n\pi t + D_n \cos\frac{3}{5}n\pi t\right)$$

$$\frac{\partial u}{\partial t}\Big|_{t=0} = \sum_{n=1}^{\infty} \sin\frac{n\pi x}{5} \times \frac{3}{5}n\pi \times \quad C_n = 0 \quad \therefore \ C_n = 0.$$

$$u = \sum_{n=1}^{\infty} \sin\frac{n\pi x}{5} \, D\cos\frac{3}{5}n\pi t$$

$$u(t=0) = \sum_{n=1}^{\infty} D_n \sin\frac{n\pi x}{5} = 4\sin(\pi x) - \sin(2\pi x) - 3\sin(5\pi x)$$

$D_5 = 4$. $D_{10} = -1$ $D_{25} = -3$ 其余 $D_n = 0$.

$$\therefore \ u = 4\sin\pi x \cos 3\pi t - \sin 2\pi x \cos 6\pi t - 3\sin 5\pi x \cos 15\pi t.$$

② 输运方程.

例：

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0 \qquad 0<x<a \qquad 0<y<b.$$

$$\left.\frac{\partial u}{\partial x}\right|_{x=0}=\frac{u_0}{a} \qquad u(a,y)=u_1+u_2\cos\frac{\pi}{b}y.$$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0}=0 \qquad \left.\frac{\partial u}{\partial y}\right|_{y=b}=0$$

解：考虑 $u=XY$. $\quad YX''+XY''=0$

$$\frac{X''}{X}=-\frac{Y''}{Y}=\lambda$$

对 $\lambda Y+Y''=0$

① 若 $\lambda=0$. $\quad Y''=0$. $\quad Y=Ay+B$. $\quad \frac{\partial u}{\partial y}=XA=0$.

平庸解.

② $\lambda<0$. 无解.

③ $\lambda>0$. $\quad Y''+\lambda Y=0$

$$\Longrightarrow \quad \lambda_n=\frac{n^2\pi^2}{b^2} \qquad Y_n(y)=\cos\frac{n\pi}{b}y \qquad \text{代回求 } X.$$

$$X_n=C_n\cosh\left(\frac{n\pi}{b}x\right)+D_n\sinh\frac{n\pi}{b}x.$$

总结：① 分离变量，考虑 $u=XY$ 代入.

② 分别解 X 与 Y

直角坐标，$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$

极坐标下，$\nabla^2 u = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial u}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \theta^2}$

球坐标，$\nabla^2 u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial u}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 u}{\partial \varphi^2}$

柱坐标，$\nabla^2 u = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial u}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2}$

本征值问题.

Legendre 方程.

形如 $(1-x^2)y'' - 2xy' + \lambda y = 0.$

求解考虑 $y = \sum_{k=1}^{\infty} a_k x^k$ 代入. 整理有递推关系.

例如：$a_{k+2} = \frac{k(k+1)-\lambda}{(k+2)(k+1)} a_k$

从而可得 a_k 与 y.

我们希望 y 不无穷大，这要求 a_k 在某处中断.

让 λ 取特定值可实现，如上面 $a_{k+2}=\frac{k(k+1)-\lambda}{(k+2)(k+1)}a_k$.

显然若 $\lambda=n(n+1)$，$n\in N$. 则当 k 取某值后后面全为0.

此时 y 中断为 n 次多项式. 将 λ 代入后即可解出 Legendre 多项式的显式.

Fourier 变换.

$$F(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\,e^{-ikx}\,dx$$

$$f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} F(k)\,e^{ikx}\,dk$$

$F(k)$ 为 $f(x)$ 的 Fourier 变换. 记

$$f(x)\longleftrightarrow F(k)$$

$$F(k)=\mathcal{F}(f(x)) \qquad f(x)=\mathcal{F}^{-1}(F(k))$$

延迟定理.

$$f(x-\xi)\longleftrightarrow e^{ik\xi}F(k).$$